ReCell — Used Phone Price Prediction

Machine Learning project · Adewale Adeyinka Falade · Python, statsmodels, scikit-learn

Problem

ReCell, a used-device marketplace, needed a data-driven way to estimate a fair price for a used phone from its specifications — to price inventory consistently and decide which devices are worth buying for resale.

Approach

- Used-phone listings with specs: screen size, cameras, memory, RAM, battery, weight, brand, 4G/5G, new price.
- Linear regression (OLS, statsmodels) with full diagnostics.
- Multicollinearity handled via VIF; tests for linearity, independence and residual normality.
- Evaluated on RMSE, MAE, R-squared, adjusted R-squared and MAPE (train and test).

Results

Metric	Train	Test
R-squared	0.847	0.833
Adjusted R-squared	0.846	0.831
MAPE	4.32%	4.51%

The model explains ~83% of price variation on unseen data with average error under 5% — strong and stable.

Key Findings

- Strongest price drivers: **screen size, main & selfie cameras, memory, RAM, weight, and normalized new price**.
- Brand matters — Nokia and Xiaomi show significant brand effects.
- 4G capability is a meaningful price driver.
- Dropping high-VIF features gave a leaner, more stable model with negligible R-squared loss.

Recommendation

ReCell should prioritise used stock with strong values on these predictive features (camera quality, memory/RAM, screen size, recent new price), since they drive both demand and resale price.

Self-authored notebook. Course brief and dataset not redistributed.